Jacob Muniko

Technology builder — web, AI, desktop, and automation

Karatina, Kenya (EAT, UTC+3) · jacobmuniko53@gmail.com · +254 727 380 378
github.com/muniko53 · muniko53.github.io/jacob-muniko-portfolio

PROFILE

I build practical software: web applications, local AI tools, Windows utilities, and automation. I start with the constraint, choose the smallest complete system, and keep going until it holds up in real use. I am in my final year of a Bachelor of Education (Arts) at Karatina University, with teaching subjects in Business Studies and Geography.

CORE CAPABILITIES

- **Web applications:** React, TypeScript, Node.js, Express, PostgreSQL, REST APIs, Socket.IO.
- **AI systems:** Gemini, OpenAI, and Groq APIs; tool calling; approval gates; model fallback; local-first assistants.
- **Desktop software:** Python, PySide6/Qt, SQLite, Windows shell APIs, threaded work, accessibility-aware interfaces.
- **Automation:** GitHub Actions, n8n, scheduled pipelines, Resend, HTML email, CSV and report generation.
- **Delivery:** Git and GitHub, Vercel, Google Cloud Run, packaging, QA, and handover notes.

SELECTED PROJECTS

Linklet — campus marketplace LIVE

A campus marketplace for Kenyan university students, with verified accounts, searchable listings, real-time messaging, offers, ratings, and admin tools.

React 19, Node.js, Express, PostgreSQL, Supabase, Socket.IO, Cloud Run, Vercel

FilePilot — Windows file intelligence BETA

A local Windows file intelligence and maintenance tool: storage analysis, exact duplicate detection, safe cleanup, diagnostics, and recovery guidance without telemetry or an account.

Python, PySide6, Qt, SQLite, SHA-256, Windows shell APIs, PyInstaller, pytest

MICA — local-first AI assistant IN DEVELOPMENT

A local-first desktop assistant for Windows that can use files, applications, a browser, memory, and scheduled tasks, with an approval gate before consequential actions.

Node.js, JavaScript, C# / .NET, SQLite, Gemini, Groq, OpenAI, Chrome DevTools Protocol, SAPI

NSE Intelligence Agent — market reporting SCHEDULED

A weekday NSE market report pipeline: one full-board price fetch, isolated news and analysis sources, a deterministic fallback when no model is available, and email delivery through Resend.

Node.js, GitHub Actions, AFX Kenya, Google News RSS, Resend, FreeLLMAPI, HTML email

TECHNICAL SKILLS

Web and product: React, TypeScript, Node.js, Express, Tailwind CSS, PostgreSQL, Supabase, REST, Socket.IO.

AI and intelligent systems: Gemini, OpenAI, Groq, tool calling, approval gates, model fallback, research and evaluation workflows.

Desktop and systems: Python, PySide6/Qt, SQLite, schema migrations, Windows shell and file APIs, threaded work, accessibility.

Automation and delivery: GitHub Actions, n8n, Resend, M-Pesa Daraja, Cloudinary, CSV and Excel reports, Vercel, Google Cloud Run, Git and GitHub.

ADDITIONAL PROJECTS

- **WordCraft One** — Microsoft Word add-in writing assistant. TypeScript, React, Office.js. In development.
- **Linklet Seller Report** — weekly seller performance email. Node.js, PostgreSQL, Resend. Completed.
- **Linklet Telegram Support Bot** — multilingual marketplace support bot. Node.js, Telegram Bot API, Groq. Completed.
- **Portfolio V3** — zero-dependency static portfolio generator. Node.js, HTML, CSS, GitHub Pages. In development.
- **Smart Sales Assistant** — WhatsApp sales assistant with an order state machine and React dashboard. Python, Flask, PostgreSQL. In development.
- **Email Summary Bot** — morning Gmail digest delivered to WhatsApp. Python, Gmail API, OpenRouter. Completed.
- **Linklet Social Poster** — daily social post approval and publishing. Node.js, Groq, Zernio. In development.

EDUCATION

Bachelor of Education (Arts)

Karatina University — School of Education and Social Sciences, Department of Curriculum Technology and Management.

Teaching subjects: Business Studies and Geography. Final year; expected to graduate around November 2027.

TEACHING PRACTICE

Mathaithi Mixed Day Senior School, Karatina — 12 weeks

Taught Grade 10 Geography and Business Studies under the Competency-Based Curriculum (CBC).

Open for freelance and contract work. Karatina, Kenya (EAT, UTC+3).